

CN MODEL PRACTICAL WRITE UP

1. IPV6 ADDRESSING FOR 8 Systems

AIM:-

To design and configure for a network consisting of 8 System using packet tracer and verify communication between the system using IPV6 ping.

PROCEDURE:-

-- PACKET TRACER --

No. of Systems: 8 PCs

HARDWARE:-

- * 8 x PC
- * 2 x Switch
- * 1 x Router

HARDWARE IPS:-

DEVICE	IPV6 Address
PC0	2001:D88:1::1/64
PC1	2001:D88:1::2/64
PC2	2001:D88:1::3/64
PC3	2001:D88:1::4/64
PC4	2001:D88:1::5/64
PC5	2001:D88:1::6/64
PC6	2001:D88:1::7/64
PC7	2001:D88:1::8/64
Router	2001:D88:1::254/64

Steps:-

- * Open Cisco Packet Tracer.
- * place 8 PCs, 2 Switches & 1 Router
- * Connect the PCs to the Switches using copper Straight-through cables.
- * Connect the Switches to the Router
- * configure IPv6 addresses on all PCs
- * Configure the Router Interface with IPv6
- * Enable IPv6 routing / Interface
- * Open the PC Command prompt
- * Test Communication using IPv6 ping

RESULT :-

Thus, IPv6 addressing was successfully configured for 8 System & Communication between the system was verified successfully.

2) BASIC VPN TUNNEL BETWEEN HEAD OFFICE AND BRANCH OFFICE AND BRANCH OFFICE

AIM:-

To configure a basic VPN tunnel between a Head office and a Branch office using CISCO packet tracer for Secure communication

PROCEDURE :-

-- PACKET TRACER --

No. of systems: 2 PCs, 2 Routers and 2 Switches

HARDWARE :-

- * 2 x PC
- * 2 x Switch
- * 2 x Router
- * Copper cables

HARDWARE IPs:-

~~Head Office~~ :- Head Office :-

Device

IP Address

PC0 192.168.1.10/24

Router0 LAN 192.168.1.1/24

Router0 WAN 16.0.0.1/30

Steps:-

- * open CISCO packet tracer
- * place two routers, two switches & two PCs
- * connect PC0 - Switch 0 → Return 0
- * connect Router 0 → Router 1
- * connect Router 1 → Switch 1 → PC1
- * configure IP addresses on both LANs

- * Configure the WAN interfaces
- * Configure routing between the two networks
- * Configure the VPN/IPsec parameters on the routers
- * Test Communication between VPN/IPsec Parameters on the Routers
- * Test Communication between PC1 & PC2

RESULT :-

Thus, a basic VPN Communication Setup between the Head office and Branch office was configured and connectivity was successfully verified.

3 WIRESHARK - RTP / VOIP PACKET ANALYSIS.

AIM

To Capture & analyze RTP/VOIP packets using Wire shark and study the study the different header fields and payload of the captured packets

PROCEDURE :

WIRESHARK :-

- * open Wireshark
- * Select the active network interface
- * Start packet capture
- * Generate or capture VOIP/RTP traffic
- * Stop The capture

RESULT

Torax route packet were successfully capture using Wireshark and the ICMP Exceeded packet header and payload were analyzed successfully.

4. Parity Bit Generation and Error checking using C

Aim

To implement parity bit generation and checking using C program for detecting errors in transmitted data

Procedure:

1. Start the C program
2. Read the binary data from the user
3. Count the number of 1s in its data
4. For even parity generation:
 - Parity bit 0 if ~~number~~ number is even
 - Parity bit 1 if number is odd
5. Append the parity bit to the data
6. Read the received data
7. Count the no. of 1s again
8. Check whether the total number of 1s is even
9. If even → No error
10. If odd → error

INPUT

Enter binary data : 1011001

OUTPUT

No. of i = 4
Even parity bit = 0

Parity checking completed.

RESULT

The C program was successfully implemented to generate and check the parity bit and parity based error detection was demonstrated successfully.